

Learning Streak & Attendance System

Overview

The Learning Streak system tracks student engagement on the SMAART Minds platform. Students must spend **30 minutes** on the dashboard daily to maintain their streak.

Current Implementation (Phase 1)

How It Works

1. **Session Start:** When a student opens the dashboard, a session begins
2. **Heartbeat:** Every 60 seconds, the system records 1 minute of activity
3. **Goal Check:** At 30 minutes, the streak counter increments
4. **Persistence:** All data saved to MongoDB in the `Avatar` collection

Data Model

```
// Avatar Collection
{
  userId: ObjectId,
  streak: Number,           // Consecutive days count
  lastActivityDate: Date,   // For streak continuity
  dailySessions: [{
    date: String,           // "YYYY-MM-DD"
    totalMinutes: Number,   // Minutes accumulated
    streakCounted: Boolean  // Has 30 min been reached?
  }],
  currentSessionStart: Date // Active session timestamp
}
```

API Endpoints

Method	Endpoint	Description
POST	<code>/api/avatar/start-session</code>	Start tracking session
POST	<code>/api/avatar/heartbeat</code>	Record 1 minute (call every 60s)
GET	<code>/api/avatar/session-status</code>	Get current session info

Frontend Components

- `useSessionTracker.js` - Hook that manages heartbeat timing
 - `LearningStreak.jsx` - Widget showing streak progress
-

Phase 2 Roadmap

2.1 Course Tracker Attendance

Track time spent on course modules and videos.

```
dailySessions: [{  
  courseMinutes: Number,  
  modulesCompleted: Number  
}]
```

2.2 Live Classes Attendance

Track attendance for coaching sessions.

```
dailySessions: [{  
  liveClassAttended: Boolean,  
  sessionId: ObjectId  
}]
```

2.3 Assessment-Based Attendance

Award streak bonus for completing assessments.

```
dailySessions: [{  
  assessmentsCompleted: Number,  
  assessmentIds: [ObjectId]  
}]
```

2.4 Community Participation

Track engagement in community discussions.

```
dailySessions: [{  
  communityPosts: Number,  
  communityComments: Number  
}]
```

Configuration

Streak Threshold

Located in `back-end/models/Avatar.js` :

```
const STREAK_THRESHOLD_MINUTES = 30; // Production: 30  
// const STREAK_THRESHOLD_MINUTES = 1; // Testing: 1
```

Heartbeat Interval

Located in `front-end/src/hooks/useSessionTracker.js` :

```
const HEARTBEAT_INTERVAL = 60000; // 60 seconds
```

Files Modified

Backend

- `models/Avatar.js` - Session tracking schema & methods
- `controllers/avatarController.js` - Session API handlers
- `routes/avatar.js` - Session routes

Frontend

- `hooks/useSessionTracker.js` - Session tracking hook
 - `components/LearningStreak.jsx` - Streak widget UI
-

Testing

1. Set `STREAK_THRESHOLD_MINUTES = 1` for quick testing
2. Login to dashboard at `http://localhost:5173/dashboard`
3. Watch the Learning Streak widget increment
4. After threshold, streak count updates with celebration
5. Reset to `30` for production